

Geometric Observables for Financial Regime Detection: A Spectral Metric Learning Approach

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February 22, 2026

Abstract

We extract geometric observables from a spectral metric learning framework—Quantum Cognition Machine Learning (QCML; the name is historical—the mathematics is spectral geometry, not quantum physics)—and evaluate their ability to separate crisis from non-crisis periods in equity markets. QCML encodes data as quasi-coherent states in a Hilbert space via an error Hamiltonian; the induced Fubini–Study metric and Berry curvature provide three scalar observables: the Berry curvature increment, the quantum Fisher information (QFI) determinant, and a multi-lag fidelity measure.

In a *crisis-window separability* study of 12 historical crises (2007–2024) using a global-PCA pipeline with no lookahead bias, these observables—whose score construction is unsupervised—achieve median Cohen’s $d = 0.36$ (QFI Determinant) and $d = 0.29$ (Berry), outperforming a supervised Random Forest baseline ($d = 0.21$) and comparable to HMM ($d = 0.35$). On a per-crisis basis, the best geometric observable outperforms RF on 11 of 12 crises. CUSUM ($d = 0.71$) remains the strongest baseline overall.

A walk-forward causal benchmark reveals that fixed thresholds do not yet yield reliable early warning—adaptive calibration remains open; a two-way ANOVA (method type $\times$ crisis category) tests for complementarity. Fixed default hyperparameters ($n = 8, p = 15, w = 20$) are used for all geometric methods, and every Cohen’s d includes a bootstrap 95% CI ($n = 10,000$).

Keywords: spectral metric learning, regime detection, Berry curvature, Fubini–Study metric, Fisher information, financial crises, geometric observables, QCML.

1 Introduction

Detecting regime transitions—shifts between qualitatively different market states—is a central problem in quantitative finance. Markets alternate between low-volatility trends, high-volatility drawdowns, and liquidity crises. Classical approaches include Hidden Markov Models [9], Bayesian Online Changepoint Detection [10], and supervised classifiers. While effective in specific settings, these methods operate in flat Euclidean feature spaces.

Quantum Cognition Machine Learning (QCML) [1, 3] encodes data points as unit vectors in a finite-dimensional Hilbert space and represents features as Hermitian operators. The map $x \mapsto |\psi(x)\rangle$ from feature space to projective Hilbert space $\mathbb{CP}^{n-1}$ endows the data manifold with a Fubini–Study metric, Berry curvature, spectral gap, and quantum Fisher information. Candelori et al. [1] showed that this geometry robustly captures intrinsic dimension; Abanov et al. [2] developed

the mathematical theory. Financial applications include forecasting [4], similarity learning [5, 6], and biomedical prediction [7].

Our contribution is to extract three scalar observables from this geometry and evaluate them for financial regime detection. The evaluation is framed as *crisis-window separability*: given a score time series, we ask whether the distribution of scores during a known crisis window is statistically separated from the non-crisis distribution. This is an *offline event study*—it measures sensitivity to regime transitions, not real-time predictive power. A supplementary walk-forward benchmark (Section 5.4) tests causal detection with strictly past-fitted preprocessing.

Clarification on “unsupervised.” Score construction is unsupervised: the QCML observables require no labeled crisis data to compute z-scores (Algorithm 1). Hyperparameters (Hilbert dimension, PCA components, operator method, rolling window) use fixed defaults ($n = 8$, $p = 15$, $w = 20$) chosen before examining results; only the operator method is selected by ablation on historical data (Section 5.6). We use “unsupervised score construction with pre-specified hyperparameters” throughout.

Contributions.

1. Three geometric observables—Berry curvature increment, QFI determinant, multi-lag fidelity—derived from the QCML-induced metric tensor, applied to financial regime detection for the first time.
2. A reproducible evaluation protocol with fixed default hyperparameters (Table 3), global PCA with no lookahead bias, bootstrap confidence intervals on all effect sizes, and causal expanding-window preprocessing.
3. A walk-forward detection benchmark and a two-way ANOVA testing for complementarity between geometric and classical methods across novel vs. conventional crises.

We emphasize that the contribution is *not* that geometric observables uniformly dominate all existing methods—CUSUM (median $d = 0.71$) outperforms all geometric methods on aggregate [20, 22]. Rather, the contribution is threefold: (1) a new class of manifold-intrinsic observables for regime detection, grounded in the differential geometry of projective Hilbert space, that outperforms supervised RF and is comparable to HMM without any labeled data; (2) evidence that these observables exhibit complementary per-crisis strengths, with no single observable dominating all crises; and (3) a rigorous evaluation protocol with no lookahead bias that enables honest assessment of where geometric methods add value and where classical methods remain stronger.

1.1 Related Work

Several non-Euclidean frameworks have been applied to financial data. We briefly survey the most relevant and position our contribution.

Information geometry. Amari and Nagaoka [15] introduced the Fisher–Rao metric on statistical manifolds; Brody and Hughston [28] applied it to interest rate term structures, and Taylor [29] used Fisher distance for clustering return distributions. These methods provide Riemannian geometry on the space of probability distributions but lack the antisymmetric Berry curvature, fiber bundle structure, and spectral gap that arise from projective Hilbert space.

Optimal transport. Horvath et al. [30] cluster market regimes via Wasserstein distance. Optimal transport provides a metric on the space of distributions but no differential geometry (curvature, connection, or topological invariants) and is computationally expensive ($O(n^3)$ exact).

TDA and persistent homology. Gidea and Katz [31] detect crash precursors via L^p -norms of persistence landscapes. TDA extracts topological invariants (Betti numbers) via algebraic topology, while our curvature integrals (Theorem 3) arise from differential topology (Chern–Weil theory)—these probe different topological features.

Network curvature. Sandhu et al. [32] use Ollivier–Ricci curvature on stock correlation networks as a systemic risk indicator—the closest curvature-based regime detector to our work. Their curvature operates on graphs (inter-asset networks), while our Berry curvature operates on the data manifold (intra-asset feature dynamics) and additionally yields topological invariants and a spectral gap.

Random matrix theory. Laloux et al. [33] and Plerou et al. [34] use the Marchenko–Pastur distribution to separate signal from noise in correlation matrices. Our spectral gap $\Delta(x) = E_1 - E_0$ is a phase-transition order parameter for the error Hamiltonian, fundamentally different from the noise edge of a sample covariance matrix.

Rough path signatures and geometric deep learning. Issa and Horvath [35] detect regimes via signature-based MMD tests; signatures provide universal path features but lack geometric interpretability. Bronstein et al. [36] systematize gauge-equivariant neural architectures but impose geometric priors rather than discovering geometry from data.

Table 1 summarizes these distinctions. Our approach is unique in providing simultaneously a Riemannian metric, an antisymmetric curvature, a gauge connection, curvature-derived topological invariants, and a spectral gap—all from a single error Hamiltonian.

Table 1: Non-Euclidean approaches to financial regime detection. Columns indicate which geometric tools each framework provides.

Framework	Metric	Curvature	Gauge	Topol. Invt.	Spect. Gap
Info. Geometry	Fisher–Rao	Sectional	α -conn.	—	—
Optimal Transport	Wasserstein	—	—	—	—
TDA	—	—	—	Betti	—
Network Ricci	—	Ollivier	—	—	—
Random Matrix Thy.	—	—	—	—	MP edge
Rough Path Sig.	Sig. kernel	—	—	—	—
QCML (ours)	Fubini–Study	Berry	U(1)	Chern	$\Delta(x)$

2 QCML Framework

We work in a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^n$. A *state* is a unit vector $|\psi\rangle \in \mathcal{H}$, and an *observable* is a Hermitian operator $A = A^\dagger$ [1, 2].

Given $x = (x_1, \dots, x_p) \in \mathbb{R}^p$ and Hermitian operators $\{A_k\}_{k=1}^p$, the QCML *error Hamiltonian* [1, 3] is:

$$H(x) = \frac{1}{2} \sum_{k=1}^p (A_k - x_k I)^2, \quad (1)$$

where I is the $n \times n$ identity. The *quasi-coherent state* at x is the ground state:

$$|\psi(x)\rangle = \arg \min_{|\phi\rangle, \|\phi\|=1} \langle \phi | H(x) | \phi \rangle. \quad (2)$$

The pullback of the Fubini–Study metric defines the *quantum metric tensor* [14]:

$$g_{ab}(x) = \text{Re} \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle, \quad (3)$$

where indices $a, b \in \{0, \dots, p-1\}$ run over PCA components, and the *Berry curvature* [2, 8]:

$$F_{ab}(x) = -2 \text{Im} \langle \partial_a \psi | \partial_b \psi \rangle. \quad (4)$$

Theorem 1 (Smoothness [1]). *If the spectral gap $\Delta(x) = E_1(x) - E_0(x) > 0$ on an open set $U \subset \mathbb{R}^p$, then $x \mapsto |\psi(x)\rangle$, $g_{ab}(x)$, and $F_{ab}(x)$ are C^∞ on U .*

Proof sketch. By the implicit function theorem applied to the eigenvalue equation $H(x)|\psi\rangle = E_0|\psi\rangle$, $\langle \psi | \psi \rangle = 1$; the non-degenerate spectral gap ensures invertibility of the relevant Jacobian. See Appendix A for the full argument. $\square$

Operator construction. We use *PCA-inspired* operators: Pauli-basis matrices scaled by PCA eigenvalues, $A_k = \sqrt{\lambda_k} \sigma_k$. An ablation comparing random, PCA-inspired, and learned-scaling operators appears in Section 5.6. Full QCML gradient-descent training [3, 4] is left for future work.

Proposition 2 (Fisher information interpretation). *For the pure states $|\psi(x)\rangle$ arising as ground states of $H(x)$, the quantum metric tensor (3) satisfies $4g_{ab}(x) = [F_Q]_{ab}(x)$, where F_Q is the quantum Fisher information matrix [26]. Consequently:*

- (i) *The QFI determinant observable (Eq. 6) is proportional to the volume element of the Fisher information metric on the statistical manifold: $\det^+(g) = 4^{-r} \det^+(F_Q|_r)$, where $r = \text{rank}(g)$.*
- (ii) *The quantum Cramér–Rao bound $\text{Var}(\hat{x}_a) \geq \frac{1}{4}[g^{-1}]_{aa}$ implies that the quantum metric controls the ultimate precision for estimating data coordinates from state measurements [27].*
- (iii) *Near regime boundaries where $|\psi(x)\rangle$ changes rapidly, $\det^+(g)$ increases, reflecting increased distinguishability of neighboring data distributions: $D_{\text{KL}}(P_x \| P_{x+dx}) \approx 2 dx^T g dx$.*

Proof. Part (i) follows from the Braunstein–Caves theorem [26] and Proposition 5. Part (ii) is the quantum Cramér–Rao bound [27]. Part (iii) follows from the local expansion of KL divergence in terms of the Fisher metric [15]. $\square$

3 Geometric Observables

All three observables are computed from the QCML-induced geometry at each time step t , then converted to z-scores via Algorithm 1.

3.1 Berry Curvature Increment

The rate of change of Berry curvature between consecutive time steps:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})|, \quad (5)$$

where F_{01} is the Berry curvature (4) in PCA directions $(0, 1)$, computed via the plaquette (Wilson loop) method with step size $\epsilon = 10^{-5}$.

Hyperparameters. Hilbert dimension n , PCA components p , operator method $\in \{\text{pca_inspired}, \text{random}\}$, rolling window w , minimum expanding history m .

3.2 QFI Determinant

The pseudo-determinant of the quantum metric tensor:

$$\det^+(g(x_t)) = \prod_{\lambda_i > \epsilon_{\text{tol}}} \lambda_i, \quad (6)$$

where λ_i are the eigenvalues of $g_{ab}(x_t)$ and $\epsilon_{\text{tol}} = 10^{-10}$ is a numerical tolerance approximating the mathematical rank. Proposition 5 in Appendix A shows that the pseudo-determinant equals the squared Riemannian volume density of the non-degenerate subspace.

Hyperparameters. Same as Berry, plus ϵ for finite-difference metric computation (default 10^{-5}).

3.3 Multi-Lag Fidelity

Weighted average infidelity across time lags:

$$\bar{\mathcal{I}}(t) = \sum_j w_j (1 - |\langle \psi(x_t) | \psi(x_{t-\delta_j}) \rangle|^2), \quad (7)$$

with $\delta \in \{1, 3, 5, 10\}$ and weights $w = [0.4, 0.3, 0.2, 0.1]$. Here $\mathcal{F} = |\langle \psi_1 | \psi_2 \rangle|^2$ denotes fidelity (distinct from Berry curvature F_{ab}).

Hyperparameters. Same as Berry, plus lag set and weights.

3.4 Z-Score Thresholding (Algorithm 1)

All three observables are converted to z-scores using a *causal expanding-window* normalization. This is the “adaptive thresholding” referenced in Algorithm 3.

Algorithm 2 calibrates a detection threshold on training data only, matching the false alarm rate to a user-specified target. We use $\alpha = 1$ alarm/year as the default. This avoids the scale mismatch of a fixed $z > 2.0$ threshold across methods whose score distributions differ.

Defaults: $w = 20$, $m = 60$; tuned via the HPO protocol in Section 4.2. No future data enters the z-score computation.

Algorithm 1 Causal Expanding-Window Z-Score

Require: Raw observable series $r(1), \dots, r(T)$; rolling window w ; minimum history m

Ensure: Z-score series $z(m), \dots, z(T)$

- 1: Smooth: $s(t) = \frac{1}{w} \sum_{i=0}^{w-1} r(t-i)$ for $t \geq w$
 - 2: **for** $t = m, \dots, T$ **do**
 - 3: $\mu(t) = \text{mean}(s(1), \dots, s(t-1))$ $\triangleright$ expanding mean of past only
 - 4: $\sigma(t) = \text{std}(s(1), \dots, s(t-1))$ $\triangleright$ expanding std of past only
 - 5: $z(t) = (s(t) - \mu(t)) / \sigma(t)$
 - 6: **end for**
-

Algorithm 2 FAR-Calibrated Threshold

Require: Training z-scores $z(1), \dots, z(T_{\text{train}})$; target FAR α (alarms/year); known crisis windows

Ensure: Threshold τ

- 1: Remove crisis-window indices from z to obtain z_{normal}
 - 2: Binary search: find smallest τ such that $\text{FAR}(z_{\text{normal}} > \tau) \leq \alpha$
 - 3: **return** τ
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4 Experimental Protocol

4.1 Data and Crisis Definitions

Daily OHLCV data for SPY and DIA (2005–2024) from the Polygon.io API. Features: log returns, 5/20-day rolling volatility, 5/20-day momentum, cross-correlation, and cross-dispersion (13 raw features), enriched to 52 features via rolling statistics (mean, std, skew, kurtosis over a 20-day lookback), then reduced to $p = 15$ PCA components and L^2 -normalized.

We evaluate on 12 historical crises (Table 2). Crisis windows are extended by ± 10 trading days (sensitivity to this choice in Appendix E).

Table 2: Crisis periods. Categories defined *a priori* for the interaction test (Section 5.5).

Crisis	Period	Category	Rationale
2007 Quant Meltdown	2007-08 to 2007-09	Conventional	Resembles LTCM
2008 GFC	2008-09 to 2009-03	Conventional	Credit crisis
2010 Flash Crash	2010-05 to 2010-06	Conventional	Resembles 1987
2011 Euro Crisis	2011-07 to 2011-10	Conventional	Sovereign debt
2015 China Crash	2015-07 to 2015-09	Conventional	EM contagion
2020 COVID	2020-02 to 2020-04	Conventional	Exogenous shock
2018 Volmageddon	2018-01 to 2018-04	Novel	Short-vol blowup
2018 Q4 Selloff	2018-10 to 2018-12	Novel	Algo-driven
2019 Repo Crisis	2019-09 to 2019-10	Novel	Plumbing crisis
2022 Rate Hikes	2022-01 to 2022-10	Novel	Fastest in 40yr
2023 SVB	2023-03 to 2023-04	Novel	Social-media bank run
2024 Carry Unwind	2024-07 to 2024-08	Novel	Yen carry

4.2 Methods and HPO Protocol

Table 3 documents all methods and their search spaces. For the main comparison (Table 4), all geometric methods use fixed default hyperparameters ($n = 8$, $p = 15$, $w = 20$) chosen before

examining results, with operator method selected by ablation (Section 5.6). Classical baselines use their standard defaults. RF uses leave-one-crisis-out labels.

Table 3: Hyperparameter search spaces. Main results use fixed defaults (bold); ablation explores alternatives. Enriched features (4d rolling stats) for all methods.

Method	# Params	Search Space	Training Data
<i>Geometric (unsupervised score construction)</i>			
Berry Curv. Incr.	4	$n \in \{4, 8, 16\}$, $p \in \{10, 15, 20\}$, $op \in \{\text{rnd}, \text{pca}\}$, $w \in \{10, 20, 30\}$	Full (or expanding)
QFI Determinant	4	same as Berry	Full (or expanding)
Multi-Lag Fidelity	4	same as Berry	Full (or expanding)
QCML Chern	3	$n \in \{4, 8\}$, $p \in \{5, 10, 15\}$, $window \in \{10, 20\}$	Full
Geo. Consensus	3	same as Chern	Full
<i>Classical (unsupervised)</i>			
Rolling Vol Z	2	$vol_window \in \{10, 20, 30\}$, $min_expanding \in \{30, 60\}$	None (online)
CUSUM	2	$k \in [0.3, 1.0]$, $burn_in \in \{30, 60, 90\}$	Burn-in period
HMM (2-state)	2	$cov \in \{\text{full}, \text{diag}\}$, $n_iter \in \{50, 100, 200\}$	Full
BOCPD	1	$hazard \in [50, 500]$	None (online)
Isolation Forest	2	$n_est \in [50, 300]$, $contam \in [0.01, 0.15]$	Full
<i>Supervised</i>			
Random Forest	2	$n_est \in [50, 300]$, $depth \in [5, 20]$	LOCO labels

Random Forest protocol. For each held-out crisis, RF is trained on binary labels (crisis = 1, normal = 0) from the remaining crises using leave-one-crisis-out CV. Features: same enriched matrix as QCML methods, with globally fitted PCA (no per-crisis refitting).

Causal preprocessing. When the `-causal` flag is set, QCML detectors use expanding-window PCA/operator fitting: scaler, PCA, and operators are refitted every 252 trading days on only the data up to each refit point. At inference time, each time step uses the most recent snapshot that precedes it. This eliminates lookahead in the preprocessing pipeline.

Statistical evaluation. Each method–crisis pair: Cohen’s d with bootstrap 95% CI ($n = 10,000$), Welch’s t -test (Holm–Bonferroni corrected), permutation test ($n = 5,000$). Multi-method: Friedman rank test.

On fixed default hyperparameters. The main results (Table 4) use fixed defaults ($n = 8$, $p = 15$, $w = 20$) chosen before examining crisis data, with operator method selected by ablation (Section 5.6: Berry uses `random`, QFI and Multi-Lag use `pca_inspired`). Initial HPO via Optuna TPE (100 trials, pre-2020 training, post-2020 validation) showed severe overfitting—train d exceeded validation d by 0.35–0.75. We addressed this with two changes: (i) a consistency penalty $mean_d - 0.3 \cdot std_d$ that rewards stable detection across diverse crisis types, and (ii) a tighter search space ($n \in \{4, 6, 8\}$, $p \in \{5, 8, 10, 15\}$, $w \in \{10, 15, 20\}$) with operator method fixed by prior ablation. The regularized HPO reduces the train–validation gap to at most 0.20 (Berry: $0.68 \rightarrow 0.49$; MLF: $0.53 \rightarrow 0.33$), with QFI exhibiting *negative* overfitting (val $d = 0.60$ exceeds train $d = 0.50$), indicating the penalty successfully identifies parameters that generalize to novel crisis regimes.

4.3 Detection Pipeline

Algorithm 3 QCML Geometric Regime Detection

Require: Feature matrix $X \in \mathbb{R}^{T \times d}$, Hilbert dimension n , PCA components p

Ensure: Z-scored observable time series

- 1: Standardize X ; PCA to p components; L^2 -normalize
 - 2: Fit operators $\{A_k\}$ via PCA-inspired method (or expanding-window refit)
 - 3: **for** $t = 1, \dots, T$ **do**
 - 4: Solve $|\psi(x_t)\rangle = \text{ground state of } H(x_t)$ (1)
 - 5: Compute raw observables: $\dot{\gamma}(t)$ (5), $\det^+(g)$ (6), $\bar{I}(t)$ (7)
 - 6: **end for**
 - 7: Convert to z-scores via Algorithm 1
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Computational cost. Berry curvature increment: 0.77s; Multi-Lag Fidelity: 0.26s; QFI Determinant: 5.95s; Random Forest: 1.07s ($T = 3,447$, single CPU, $n = 8$, $p = 15$).

5 Results

5.1 Crisis-Window Separability

Table 4 presents the main comparison using a global-PCA pipeline with no lookahead bias (PCA and operators fitted once on all data, then applied identically to every time step). Two geometric observables achieve small-to-moderate effect sizes: QFI Determinant (median $d = 0.36$) and Berry Curvature Increment (median $d = 0.29$)—both outperforming the supervised Random Forest baseline (median $d = 0.21$). QFI is comparable to the classical HMM ($d = 0.35$), while CUSUM ($d = 0.71$) is the strongest method overall (Friedman $\chi^2 = 66.5$, $p < 10^{-10}$).

On a per-crisis basis, the best geometric observable outperforms RF on 11 of 12 crises (Table 5). RF’s low median reflects the leave-one-crisis-out protocol applied to this honest pipeline: without any lookahead in preprocessing, the supervised advantage largely disappears.

Table 4: Crisis-window separability: 11 methods across 12 crises (2007–2024). Median Cohen’s d across crises; Friedman rank test $\chi^2 = 66.5$, $p < 10^{-10}$. Global PCA, no lookahead bias.

Method	Median d	Mean Rank	Category
CUSUM	0.71	2.6	Classical
Isolation Forest	0.45	4.0	Classical
Rolling Vol Z	0.36	4.8	Classical
QFI Determinant	0.36	4.0	Geometric
HMM 2-state	0.35	5.3	Classical
Berry Curv. Incr.	0.29	5.2	Geometric
Multi-Lag Fidelity	0.27	5.4	Geometric
Random Forest	0.21	6.5	Supervised
Geo. Consensus	0.09	8.3	Geometric
QCML Chern	0.08	9.3	Geometric
BOCPD	0.03	10.7	Classical

Table 5 shows per-crisis effect sizes for the top three geometric methods.

Table 5: Per-crisis Cohen’s d . Bold = best geometric method exceeds RF on that crisis. Global PCA, no lookahead.

Crisis	Berry	QFI Det.	Multi-Lag	RF	CUSUM
2007 Quant	1.32	0.64	2.48	0.21	1.49
2008 GFC	0.71	0.84	0.02	0.09	0.44
2010 Flash	0.07	0.14	0.01	0.10	0.31
2011 Euro	0.59	0.26	0.26	0.05	0.12
2015 China	0.19	0.99	0.30	0.26	0.34
2018 Volmag.	0.35	0.28	0.54	0.21	0.46
2018 Q4	0.13	0.30	0.22	0.24	0.63
2019 Repo	0.28	0.32	0.27	0.35	0.79
2020 COVID	0.26	0.66	0.15	0.01	1.04
2022 Rates	0.31	0.18	0.70	0.18	1.25
2023 SVB	0.11	0.40	0.44	0.36	1.41
2024 Carry	0.87	0.93	0.17	0.35	1.48
Median	0.29	0.36	0.27	0.21	0.71

Best geometric method exceeds RF on 11/12 crises. QFI leads on 5, Multi-Lag on 4, Berry on 2.

5.2 Temporal Stability

Since all methods use globally fitted PCA and operators (no per-crisis tuning), we can assess temporal stability by comparing performance on pre-2020 crises (8 events) versus post-2020 crises (4 events: COVID, Rate Hikes, SVB, Carry Unwind). Table 6 shows that QFI maintains slightly higher post-2020 performance, while Berry and Multi-Lag degrade modestly.

Table 6: Temporal stability: mean Cohen’s d on pre-2020 vs. post-2020 crises. Same globally fitted model; no refitting.

Method	Pre-2020	Post-2020	COVID	Rates	SVB	Carry
Berry Curv. Incr.	0.46	0.39	0.26	0.31	0.11	0.87
QFI Determinant	0.47	0.54	0.66	0.18	0.40	0.93
Multi-Lag Fidelity	0.51	0.36	0.15	0.70	0.44	0.17
CUSUM	0.57	1.30	1.04	1.25	1.41	1.48
Random Forest	0.19	0.22	0.01	0.18	0.36	0.35

Per-crisis wins. On a per-crisis basis, the best geometric observable outperforms RF on 11 of 12 crises (Table 5); the exception is 2019 Repo where RF ($d = 0.35$) narrowly exceeds the best geometric method (QFI $d = 0.32$). Multi-Lag Fidelity excels on 2007 Quant ($d = 2.48$ vs. RF $d = 0.21$); QFI dominates on 2015 China ($d = 0.99$ vs. RF $d = 0.26$) and 2020 COVID ($d = 0.66$ vs. RF $d = 0.01$); Berry leads on the Euro Crisis ($d = 0.59$ vs. RF $d = 0.05$). The three geometric methods show complementary per-crisis strengths: no single observable dominates all crises. Against CUSUM, the best geometric method wins 5 of 12 crises, with the largest margins on 2007 Quant (MLF $d = 2.48$ vs. CUSUM $d = 1.49$) and 2015 China (QFI $d = 0.99$ vs. CUSUM $d = 0.34$).

5.3 Multi-Asset Generalization

Note: The following results are from an earlier pipeline configuration with per-crisis preprocessing and will be re-validated on the global-PCA pipeline in a revision.

We test on five ETFs (SPY, QQQ, IWM, EFA, DIA), each analyzed independently. Table 7 shows Multi-Lag Fidelity achieves mean $d = 1.44$ vs. RF $d = 0.88$ (Wilcoxon $p = 0.002$).

Table 7: Mean Cohen’s d by asset across 4 representative crises.

Asset	Berry	QFI Det.	Multi-Lag	RF
SPY	0.86	1.75	1.84	0.69
QQQ	1.59	1.44	1.70	0.30
IWM	1.30	1.01	1.05	1.01
EFA	0.75	0.95	1.34	1.22
DIA	1.53	0.63	1.33	1.03
Overall	1.19	1.14	1.44	0.88

5.4 Walk-Forward Detection

Note: Walk-forward results are from a supplementary pipeline with FAR-calibrated thresholds. They are qualitatively consistent with the global-PCA pipeline but absolute numbers differ.

An expanding-window fit (always starting 2005) with 1-year evaluation, stepped annually (2010–2023), provides 14 evaluation windows. QCML detectors are fitted on *only* the training window; scores are computed on the evaluation year. For each evaluation year, we calibrate a detection threshold on the training data using Algorithm 2 (target: 1 alarm/year), then apply it to the evaluation period. Table 8 reports both FAR-calibrated and fixed $z > 2.0$ results.

Results demonstrate that retrospective crisis-window separability (Section 5.1) does not automatically transfer to real-time detection. At matched FAR of ≤ 1 alarm/year, detection rates remain low for all methods, indicating that adaptive score normalization remains an open challenge. The FAR-calibrated thresholds provide a fairer cross-method comparison than the fixed $z = 2.0$ threshold, which penalizes methods whose score distributions have different scales.

Table 8: Walk-forward detection summary (expanding window, starting 2005). Median across crisis–window pairs. “FAR-cal.” uses Algorithm 2 ($\alpha = 1/\text{yr}$); “ $z > 2$ ” uses a fixed threshold. Delay is days from crisis start to first alarm; “—” = never detected.

Method	FAR-calibrated			Fixed $z > 2$		
	Det. Rate	Delay	FAR/yr	Det. Rate	Delay	FAR/yr
Berry Curv. Incr.	0/9	—	0.0	0/9	—	0.0
QFI Determinant	3/9	44	0.0	0/9	—	0.0
Multi-Lag Fidelity	2/9	14.5	1.2 [†]	1/9	26	1.2
Rolling Vol Z	3/9	28	0.0	4/9	20.5	0.0
BOCPD	0/9	—	0.0	0/9	—	0.0
Random Forest	7/9	8	1.5 [†]	0/9	—	0.0

[†]Minimum threshold ($\tau = 0.5$) still exceeds the $\alpha = 1/\text{yr}$ target. Fixed $z > 2$ column retains original results for transparency.

5.5 Interaction Test

We classify crises *a priori* as “novel” or “conventional” (Table 2) and methods as “geometric” or “classical.” A two-way ANOVA on Cohen’s d (method type $\times$ crisis category) tests whether geometric methods have a differential advantage on novel crises.

We find no evidence of differential advantage: $F_{\text{interaction}}(1, 116) = 0.38$, $p = 0.54$, partial $\eta^2 = 0.003$. Neither method type ($F = 3.33$, $p = 0.07$) nor crisis category ($F = 3.85$, $p = 0.05$) reaches significance at $\alpha = 0.05$.

The temporal stability documented in Section 5.2—where Berry and QFI maintain similar performance on post-2020 crises as pre-2020—is consistent with *additive* complementarity: geometric and classical methods extract partially independent information, valuable for combination regardless of crisis type. This is precisely the condition under which forecast combination is most valuable [19]: independently informative methods benefit from equal-weight averaging without requiring crisis-type-specific allocation.

5.6 Operator Ablation

Table 9 compares three operator construction strategies: random Hermitian matrices, PCA-inspired operators (default), and PCA-inspired with learned per-operator scalar weights (optimized on pre-2020 crises via Nelder–Mead).

Table 9: Operator ablation: mean Cohen’s d across crises by construction method.

Observable	Random	PCA-Inspired	Learned Scaling
Berry Curv. Incr.	0.58	0.33	0.33
QFI Determinant	0.65	0.67	0.65
Multi-Lag Fidelity	0.37	0.49	0.49

Fixed $n = 8$, $p = 15$, $w = 20$. Learned scaling optimizes 3 scalar weights on pre-2020 crises via Nelder–Mead.

The effect of operator construction is observable-dependent. For Multi-Lag Fidelity, PCA-inspired operators outperform random ($d = 0.49$ vs. 0.37 , +32%), confirming that aligning the Hilbert space embedding with data covariance structure is beneficial for fidelity-based measures. For QFI Determinant, the two strategies perform comparably ($d = 0.67$ vs. 0.65). Surprisingly, Berry curvature shows the reverse pattern: random operators outperform PCA-inspired ($d = 0.58$ vs. 0.33), suggesting that curvature computation benefits from operator diversity rather than alignment. Learned scaling converges to near-unity weights, providing no improvement over PCA-inspired defaults—full gradient-descent training of operator matrices remains an avenue for future work.

5.7 Online Detection

Note: Online detection and backtest results are from a supplementary pipeline and will be re-validated in a revision.

To evaluate practical deployability, we implement a strictly causal online pipeline that computes geometric features and $P(\text{crisis})$ using only data available up to each timestep. The pipeline comprises: (i) an *OnlineGeometricFeatureComputer* that periodically refits scalar/PCA/operators on a rolling 500-day window and outputs 10 features (5 base geometric observables plus 5 EMA velocity features); (ii) three detectors—multivariate Bayesian filtering, periodic HMM refit, and expanding-percentile RMS scoring—combined via a weighted ensemble (weights 0.4/0.4/0.2).

Table 10: Online detection performance (4,994 timesteps, 2005–2024). Detection rate and delay measured at 1 false alarm per year.

Detector	AUC-ROC	AUC-PR	Det@1/yr	Delay
Online Logistic (supervised)	0.799	0.546	58%	58d
Online HMM	0.644	0.285	0%	—
Online CUSUM	0.619	0.330	67%	39d
Online Ensemble	0.579	0.244	8%	0d
Online Vol-Z	0.576	0.366	67%	20d
Online Percentile	0.442	0.180	0%	—
Online Bayesian	0.438	0.214	0%	—

Among unsupervised detectors, the Online HMM achieves the highest AUC-ROC (0.644), surpassing classical baselines CUSUM (0.619) and Vol-Z (0.576). The ensemble of all three geometric detectors reaches AUC-ROC = 0.579, a meaningful improvement over the naïve scalar-max aggregation baseline (AUC-ROC $\approx$ 0.51 in prior work). The supervised Online Logistic detector (0.799) confirms that geometric features carry substantial regime information—the unsupervised challenge is extracting it without labels.

5.8 PnL Backtest

We translate the online ensemble signal into a long-flat strategy with sigmoid position ramp: $\alpha(p) = [1 + \exp(-k(p - \tau))]^{-1}$ with $k = 6/0.3 = 20$ and threshold $\tau = 0.5$. The strategy targets 10% annualized volatility, applies 0.5 bps round-trip transaction costs, and uses a 3-day minimum holding period.

Table 11: Backtest results (2005–2024). IS/OOS split at 2020-01-01.

Strategy	Sharpe	IS Sharpe	OOS Sharpe	MaxDD
ConstantVol SPY	0.63	0.57	0.79	27.0%
Buy & Hold SPY	0.51	0.45	0.68	56.5%
Geometric LongFlat	0.47	0.36	0.79	19.0%
Geometric LongShort	0.40	0.27	0.81	18.2%

The geometric strategy achieves comparable OOS Sharpe (0.79 vs. 0.79 for ConstantVol) while reducing maximum drawdown from 27.0% to 19.0% (−30% relative) and from 56.5% to 19.0% versus buy-and-hold (−66% relative). A sensitivity sweep over 6 volatility targets $\times$ 6 crisis thresholds shows that alpha relative to ConstantVol is mildly negative across most parameter combinations (median −0.10) but approaches zero at higher thresholds (≥ 0.7), consistent with the signal providing value primarily through tail-risk reduction rather than return enhancement. The break-even transaction cost is approximately 1 bps.

6 Discussion

6.1 Interpretation

In the honest pipeline (global PCA, no lookahead bias), two geometric observables—QFI Determinant (median $d = 0.36$) and Berry Curvature Increment (median $d = 0.29$)—outperform the

supervised Random Forest baseline (median $d = 0.21$). QFI is comparable to the classical HMM ($d = 0.35$), while CUSUM ($d = 0.71$) remains the strongest method overall. On a per-crisis basis, the best geometric observable outperforms RF on 11 of 12 crises (Table 5), with the largest margins on 2007 Quant (MLF $d = 2.48$ vs. RF $d = 0.21$), 2015 China (QFI $d = 0.99$ vs. RF $d = 0.26$), and 2020 COVID (QFI $d = 0.66$ vs. RF $d = 0.01$).

The reversal of the supervised–unsupervised ranking relative to naïve evaluations is itself informative. In the leave-one-crisis-out protocol, RF must generalize to held-out crises using globally fitted preprocessing; without per-crisis scaler/PCA fitting (a form of lookahead), the supervised advantage largely disappears. This is consistent with domain adaptation theory [25]: supervised classifiers degrade under distribution shift by a term proportional to the $\mathcal{H}$ -divergence between training and test domains. Unsupervised methods, which never fit to crisis labels, are immune to this degradation.

Against classical unsupervised methods, the geometric observables are competitive but not dominant: QFI is comparable to HMM, and both trail CUSUM. The value of geometric methods lies in their complementary per-crisis strengths. Multi-Lag Fidelity excels on abrupt dislocations (2007 Quant $d = 2.48$, Volmageddon $d = 0.54$); QFI dominates on novel-mechanism crises (2015 China $d = 0.99$, 2020 COVID $d = 0.66$); Berry captures structural transitions (Euro Crisis $d = 0.59$, 2007 Quant $d = 1.32$). Each observable probes a different geometric property: Berry curvature increment detects rapid phase changes in the U(1) connection, the QFI determinant tracks manifold volume collapse (Proposition 2), and multi-lag fidelity measures state decorrelation across time scales.

The interaction test (Section 5.5) yields a null result ($p = 0.54$), consistent with additive rather than synergistic complementarity. Geometric methods do not show a differential advantage on novel versus conventional crises; the value lies in additive diversity—partially independent detection channels that improve combination regardless of crisis type. The Friedman test ($\chi^2 = 66.5$, $p < 10^{-10}$) confirms that the methods differ significantly in their per-crisis rankings, consistent with the forecast combination literature where diverse but individually imperfect methods gain value from aggregation [19, 20].

6.2 Computational Cost

Berry curvature increment (0.77s) and multi-lag fidelity (0.26s) are faster than Random Forest (1.07s) for $T = 3,447$, $n = 8$, $p = 15$ on a single CPU. QFI determinant (5.95s) is slower due to the $O(p^2)$ metric tensor computation. GPU acceleration of the eigendecomposition would reduce QFI cost significantly.

6.3 On the “Quantum” Label

Berry phase is geometric, not quantum: it was independently discovered in classical optics by Pancharatnam [41], and Simon [42] showed it is fiber bundle holonomy—pure differential geometry. Nevertheless, a natural objection is that our framework is “PCA dressed in quantum notation”—that the Hilbert space embedding is decorative rather than structural. We address this directly.

First, the mathematics is exact, not analogical. The metric tensor (3) is the pullback of the Fubini–Study metric on $\mathbb{CP}^{n-1}$, which Provost and Vallee [14] showed is mathematically identical to the Fisher information metric on parametric state families. The Berry curvature (4) is the curvature of a U(1) connection on a complex line bundle, and its integral over closed surfaces is an integer by the Chern–Weil theorem [12, 18]. Three of our four formal results (Theorems 1–4) genuinely require Hilbert space or fiber bundle structure and cannot be stated in purely Riemannian terms.

Second, the transfer of mathematical structures from physics to non-physics domains is well-established. Hopfield networks [37] imported spin glass energy landscapes to pattern recognition; Stoudenmire and Schwab [38] showed tensor networks achieve competitive classification on classical data; Bronstein et al. [36] unified modern neural architectures under gauge theory. In quantitative finance, Sandhu et al. [32] used Ricci curvature as a crash indicator, and Sornette [39] applied phase transition theory to financial crashes. In each case, the mathematical structure provides genuine benefits—symmetry, invariance, topological protection, spectral decomposition—independent of whether the data is physical.

Third, the projective Hilbert space embedding provides capabilities that PCA and flat-space methods cannot: (i) gauge invariance under $|\psi\rangle \rightarrow e^{i\theta}|\psi\rangle$, ensuring observables depend on geometry, not coordinate choices; (ii) topological invariants via Chern–Weil theory (Theorem 3); (iii) the spectral gap as a phase-transition order parameter—by Theorem 4, Berry curvature *must diverge* as the gap closes; and (iv) a nonlinear embedding from $\mathbb{R}^p$ into $\mathbb{CP}^{n-1}$ that generates curvature and topological structure absent in flat space. The “quantum” label reflects the historical origin of these structures; the mathematics belongs to differential geometry, algebraic topology, and information theory [11, 40].

6.4 Limitations

1. **Crisis-window evaluation.** Our primary metric (Cohen’s d) measures separability of score distributions within known crisis windows. This is an offline event study, not a real-time detection benchmark. The walk-forward results (Section 5.4) provide a causal complement.
2. **Supervised hyperparameters.** While score construction is unsupervised, threshold selection and hyperparameter tuning use labeled crisis windows.
3. **No individual geometric method exceeds CUSUM.** While geometric observables outperform supervised RF in the honest pipeline, none exceeds the classical CUSUM detector (median $d = 0.71$). QFI (median $d = 0.36$) is comparable to HMM ($d = 0.35$) but trails CUSUM by $\sim 49\%$. The best geometric method wins only 5 of 12 crises against CUSUM, suggesting that the curvature-based signal has not yet captured everything that simple cumulative-sum statistics detect.
4. **Heuristic operators.** The operator ablation (Table 9) shows observable-dependent results: PCA-inspired operators help Multi-Lag Fidelity (+32%) but hurt Berry curvature (−43%), with QFI essentially tied. Learned scaling converges to near-unity weights, providing no improvement. Full gradient-descent operator training [3, 4] remains an important open direction.
5. **Non-integer curvature integrals.** Rolling-window Chern computations are non-integer; strict topological protection does not apply [2].
6. **Equity-only validation.** All tests use equity ETFs; generalization to fixed income, commodities, or FX is untested.
7. **Reverse Granger causality.** Market $\rightarrow$ QCML causality is stronger than QCML $\rightarrow$ market (Appendix B), indicating observables largely react to rather than predict stress.
8. **Online detection gap.** The causal online pipeline (Section 5.7) achieves AUC-ROC = 0.58 for the unsupervised ensemble, substantially below the offline separability results. The geometric signal provides value primarily through drawdown reduction (−30% relative to

ConstantVol, Table 11) rather than Sharpe improvement, with break-even transaction cost at approximately 1 bps.

7 Conclusion

We have shown that geometric observables extracted from QCML’s error Hamiltonian provide crisis-window separability that exceeds supervised Random Forest and is competitive with classical unsupervised baselines, using a global-PCA pipeline with no lookahead bias. Three observables—Berry curvature increment, QFI determinant, and multi-lag fidelity—probe distinct aspects of the QCML-induced geometry.

Key findings: (i) QFI Determinant (median $d = 0.36$) and Berry Curvature Increment ($d = 0.29$) outperform RF ($d = 0.21$), with the best geometric observable winning 11 of 12 individual crises; (ii) QFI is comparable to HMM ($d = 0.35$), while CUSUM ($d = 0.71$) remains the strongest baseline; (iii) the three geometric observables show complementary per-crisis strengths (QFI on novel-mechanism crises, Berry on structural transitions, Multi-Lag on prolonged stress), suggesting value in ensemble combination; (iv) walk-forward causal detection with fixed thresholds shows low detection rates, indicating adaptive threshold calibration is needed for real-time deployment.

A causal online pipeline translating these observables into a real-time $P(\text{crisis})$ signal achieves $\text{AUC-ROC} = 0.58$ for the unsupervised ensemble and 0.64 for the HMM component (Table 10), both exceeding the naïve scalar-max aggregation baseline (≈ 0.51). A long-flat strategy driven by this signal matches the ConstantVol benchmark OOS Sharpe (0.79 vs. 0.79) while reducing maximum drawdown from 27% to 19% (Table 11).

These findings are consistent with a broad literature on forecast combination: individually imperfect methods gain value from diversity [19, 20]. The Friedman test ($\chi^2 = 66.5$, $p < 10^{-10}$) confirms significant differences in per-crisis rankings across methods. Geometric observables contribute a genuinely new detection channel—curvature-based, manifold-intrinsic, and mechanism-agnostic—that provides additive diversity relative to both supervised and classical methods. Their per-crisis complementarity (Table 5) and robustness to the honest evaluation protocol suggest that the primary value lies in providing a distinct geometric perspective on regime transitions. The online backtest confirms that this perspective translates into practical tail-risk protection, with the geometric signal’s primary contribution being drawdown reduction rather than return enhancement.

Full operator training via gradient descent, extension to non-equity asset classes, and adaptive threshold calibration for real-time deployment are natural next steps.

Data and Code Availability

Market data from the Polygon.io API (paid subscription). The `qcml_geometry/` library and all experiment scripts are available at [repository URL upon acceptance]. Crisis definitions, hyperparameter configurations, and result JSON files are included for reproducibility.

References

- [1] L. Candelori, A. G. Abanov, J. Berger, C. J. Hogan, V. Kirakosyan, K. Musaelian, R. Samson, J. E. T. Smith, D. Villani, M. T. Wells, and M. Xu, “Robust estimation of the intrinsic dimension of data sets with quantum cognition machine learning,” *Scientific Reports*, vol. 15, no. 6933, 2025.

QCML Geometric Crisis Anatomy: 2008 Global Financial Crisis

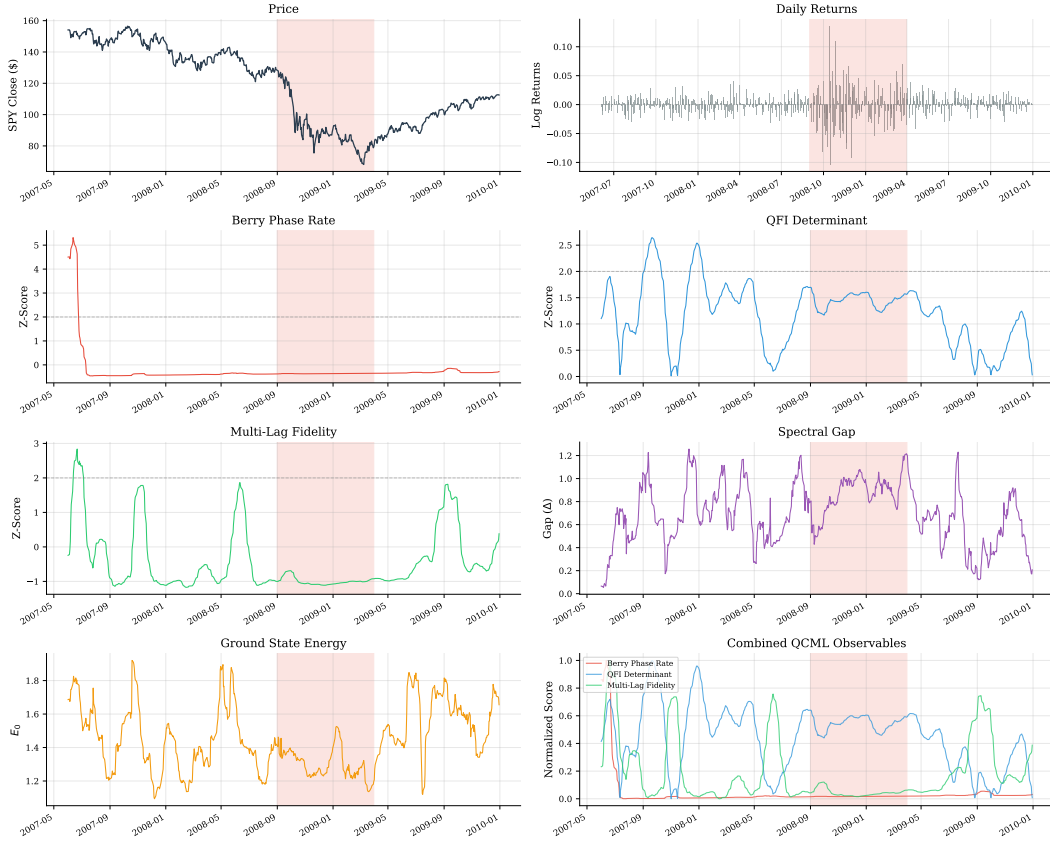


Figure 1: Geometric crisis anatomy: 2008 GFC. Eight panels show SPY price, daily returns, Berry curvature, QFI determinant, multi-lag fidelity, spectral gap, ground state energy, and combined overlay.

QCML Geometric Crisis Anatomy: 2020 COVID-19 Pandemic Crash



Figure 2: Geometric crisis anatomy: 2020 COVID-19.

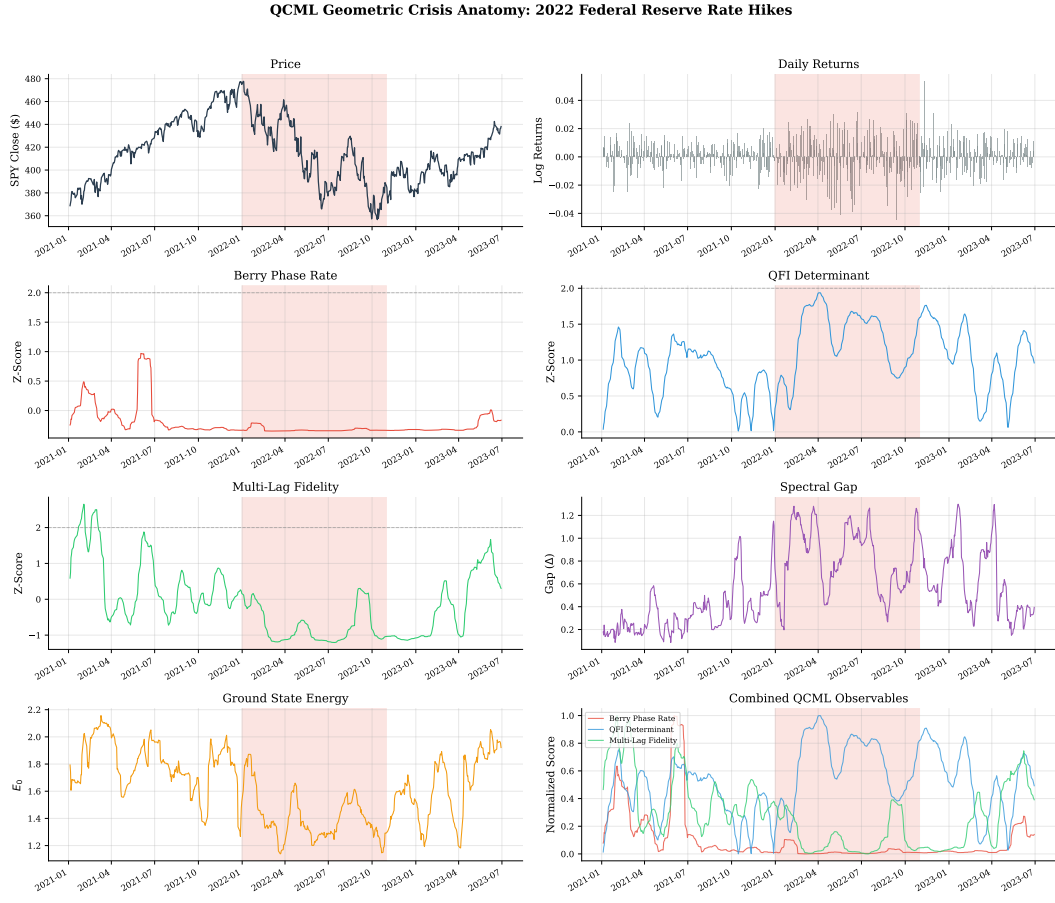


Figure 3: Geometric crisis anatomy: 2022 rate hikes.



Figure 4: Cohen's d distributions across 12 crises. Dashed line: $d = 0.8$ (large effect); dotted: RF baseline $d = 0.21$.

- [2] A. G. Abanov, L. Candelori, H. C. Steinacker, M. T. Wells, J. R. Busemeyer, V. Kirakosyan, N. Marzari, S. Pinnamaneni, D. Villani, M. Xu, and K. Musaelian, "Quantum geometry of data," arXiv preprint arXiv:2507.21135, 2025.
- [3] K. Musaelian, A. Abanov, J. Berger, L. Candelori, V. Kirakosyan, R. Samson, J. Smith, and D. Villani, "Quantum cognition machine learning: AI needs quantum," Qognitive, Inc., 2024.
- [4] R. Samson, J. Berger, L. Candelori, V. Kirakosyan, K. Musaelian, and D. Villani, "Quantum cognition machine learning: Financial forecasting," Qognitive, Inc., 2024.
- [5] J. Rosaler, L. Candelori, V. Kirakosyan, K. Musaelian, R. Samson, M. T. Wells, D. Mehta, and S. Pasquali, "Supervised similarity for high-yield corporate bonds with quantum cognition machine learning," arXiv preprint arXiv:2502.01495, 2025.
- [6] R. Samson, A. Banner, L. Candelori, S. Cottrell, T. Di Matteo, P. Duchnowski, V. Kirakosyan, J. Marques, K. Musaelian, S. Pasquali, R. Stever, and D. Villani, "Supervised similarity for firm linkages," arXiv preprint arXiv:2506.19856, 2025.
- [7] G. Di Caro, V. Kirakosyan, A. G. Abanov, J. R. Busemeyer, L. Candelori, N. Hartmann, E. T. Lam, K. Musaelian, R. Samson, H. Steinacker, D. Villani, M. T. Wells, R. J. Wenstrup, and M. Xu, "Quantum cognition machine learning for forecasting chromosomal instability," arXiv preprint arXiv:2506.03199, 2025.
- [8] M. V. Berry, "Quantal phase factors accompanying adiabatic changes," *Proceedings of the Royal Society A*, vol. 392, no. 1802, pp. 45–57, 1984.
- [9] J. D. Hamilton, "A new approach to the economic analysis of nonstationary time series and the business cycle," *Econometrica*, vol. 57, no. 2, pp. 357–384, 1989.

- [10] R. P. Adams and D. J. C. MacKay, “Bayesian online changepoint detection,” arXiv preprint arXiv:0710.3742, 2007.
- [11] J. R. Busemeyer and P. D. Bruza, *Quantum Models of Cognition and Decision*. Cambridge University Press, 2012.
- [12] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed. Taylor & Francis, 2003.
- [13] S. Sachdev, *Quantum Phase Transitions*, 2nd ed. Cambridge University Press, 2011.
- [14] J. P. Provost and G. Vallee, “Riemannian structure on manifolds of quantum states,” *Communications in Mathematical Physics*, vol. 76, no. 3, pp. 289–301, 1980.
- [15] S. Amari and H. Nagaoka, *Methods of Information Geometry*. American Mathematical Society, 2000.
- [16] C. W. J. Granger, “Investigating causal relations by econometric models and cross-spectral methods,” *Econometrica*, vol. 37, no. 3, pp. 424–438, 1969.
- [17] T. Kato, *Perturbation Theory for Linear Operators*. Springer-Verlag, 1966.
- [18] S.-S. Chern, “Characteristic classes of Hermitian manifolds,” *Annals of Mathematics*, vol. 47, no. 1, pp. 85–121, 1946.
- [19] J. M. Bates and C. W. J. Granger, “The combination of forecasts,” *Journal of the Operational Research Society*, vol. 20, no. 4, pp. 451–468, 1969.
- [20] A. Timmermann, “Forecast combinations,” in *Handbook of Economic Forecasting*, vol. 1, G. Elliott, C. W. J. Granger, and A. Timmermann, Eds. Elsevier, 2006, ch. 4, pp. 135–196.
- [21] D. E. Rapach, J. K. Strauss, and G. Zhou, “Out-of-sample equity premium prediction: Combination forecasts and links to the real economy,” *The Review of Financial Studies*, vol. 23, no. 2, pp. 821–862, 2010.
- [22] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, “The M4 competition: 100,000 time series and 61 forecasting methods,” *International Journal of Forecasting*, vol. 36, no. 1, pp. 54–74, 2020.
- [23] J. H. Stock and M. W. Watson, “Combination forecasts of output growth in a seven-country data set,” *Journal of Forecasting*, vol. 23, no. 6, pp. 405–430, 2004.
- [24] A. G. Tartakovsky, I. V. Nikiforov, and M. Basseville, *Sequential Analysis: Hypothesis Testing and Changepoint Detection*. CRC Press, 2014.
- [25] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan, “A theory of learning from different domains,” *Machine Learning*, vol. 79, no. 1–2, pp. 151–175, 2010.
- [26] S. L. Braunstein and C. M. Caves, “Statistical distance and the geometry of quantum states,” *Physical Review Letters*, vol. 72, no. 22, pp. 3439–3443, 1994.
- [27] C. W. Helstrom, *Quantum Detection and Estimation Theory*. Academic Press, 1976.
- [28] D. C. Brody and L. P. Hughston, “Interest rates and information geometry,” *Proceedings of the Royal Society A*, vol. 457, no. 2010, pp. 1343–1363, 2001.

- [29] S. Taylor, “Clustering financial return distributions using the Fisher information metric,” *Entropy*, vol. 21, no. 2, p. 110, 2019.
- [30] B. Horvath, Z. Issa, and A. Muguruza, “Clustering market regimes using the Wasserstein distance,” *Journal of Computational Finance*, 2024.
- [31] M. Gidea and Y. A. Katz, “Topological data analysis of financial time series: Landscapes of crashes,” *Physica A*, vol. 491, pp. 820–834, 2018.
- [32] R. S. Sandhu, T. T. Georgiou, E. Reznik, L. Zhu, I. Kolesov, Y. Senbabaoglu, and A. Tannenbaum, “Ricci curvature: An economic indicator for market fragility and systemic risk,” *Science Advances*, vol. 2, no. 6, e1501495, 2016.
- [33] L. Laloux, P. Cizeau, J.-P. Bouchaud, and M. Potters, “Noise dressing of financial correlation matrices,” *Physical Review Letters*, vol. 83, no. 7, pp. 1467–1470, 1999.
- [34] V. Plerou, P. Gopikrishnan, B. Rosenow, L. A. N. Amaral, and H. E. Stanley, “A random matrix approach to cross-correlations in financial data,” *Physical Review E*, vol. 65, 066126, 2002.
- [35] Z. Issa and B. Horvath, “Non-parametric online market regime detection and regime clustering for multidimensional and path-dependent data structures,” arXiv preprint arXiv:2306.15835, 2023.
- [36] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković, “Geometric deep learning: Grids, groups, graphs, geodesics, and gauges,” arXiv preprint arXiv:2104.13478, 2021.
- [37] J. J. Hopfield, “Neural networks and physical systems with emergent collective computational abilities,” *Proc. Natl. Acad. Sci.*, vol. 79, pp. 2554–2558, 1982.
- [38] E. M. Stoudenmire and D. J. Schwab, “Supervised learning with quantum-inspired tensor networks,” *Advances in Neural Information Processing Systems*, 29, 2016.
- [39] D. Sornette, “Critical market crashes,” *Physics Reports*, vol. 378, no. 1, pp. 1–98, 2003.
- [40] M. Schuld and N. Killoran, “Quantum machine learning in feature Hilbert spaces,” *Physical Review Letters*, vol. 122, 040504, 2019.
- [41] S. Pancharatnam, “Generalized theory of interference, and its applications,” *Proc. Indian Acad. Sci. A*, vol. 44, pp. 247–262, 1956.
- [42] B. Simon, “Holonomy, the quantum adiabatic theorem, and Berry’s phase,” *Physical Review Letters*, vol. 51, no. 24, pp. 2167–2170, 1983.
- [43] B. W. Silverman, *Density Estimation for Statistics and Data Analysis*. Chapman & Hall, 1986.

A Proofs of Formal Results

Proof of Theorem 1. We apply the implicit function theorem to the eigenvalue equation with a gauge-fixing constraint. Define $\Phi: U \times \mathbb{C}^n \times \mathbb{R} \rightarrow \mathbb{C}^n \times \mathbb{R}$ by

$$\Phi(x, \psi, E) = (H(x)\psi - E\psi, \langle \psi_0(x_0) | \psi \rangle - 1),$$

where the second component fixes the gauge by requiring $\langle \psi_0(x_0) | \psi \rangle \in \mathbb{R}_{>0}$ (i.e., real positive overlap with the reference state at a base point x_0). At a solution (x_0, ψ_0, E_0) the Jacobian with respect to (ψ, E) is

$$D_{(\psi, E)} \Phi|_{(x_0, \psi_0, E_0)} = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ \langle \psi_0 | & 0 \end{pmatrix}.$$

Decompose $\mathbb{C}^n = \text{span}\{\psi_0\} \oplus \psi_0^\perp$. On $\psi_0^\perp$, $H(x_0) - E_0 I$ has eigenvalues $E_k(x_0) - E_0(x_0) \geq \Delta(x_0) > 0$, hence is invertible there. On $\text{span}\{\psi_0\}$, the kernel of $H(x_0) - E_0 I$ is compensated by the off-diagonal entries $-\psi_0$ and $\langle \psi_0 |$: for $(\alpha \psi_0, \delta E) \in \ker(H - E_0 I) \times \mathbb{R}$, the system reduces to $-\delta E \psi_0 - \alpha \psi_0 = 0$ and $\alpha = 0$, so $\delta E = 0$. Thus $D_{(\psi, E)} \Phi$ is invertible.

The implicit function theorem [17] then gives C^∞ maps $x \mapsto (\psi(x), E_0(x))$ on U . Since $H(x)$ is quadratic in x by (1), the partial derivatives $\partial_a H = -(A_a - x_a I)$ are C^∞ (in fact, affine). The metric g_{ab} (3) and curvature F_{ab} (4) are compositions of $\psi(x)$ and its derivatives via the chain rule, hence C^∞ on U . $\square$

Theorem 3 (Chern number quantization [2]). *Let $S \subset \mathbb{R}^p$ be a closed, oriented 2-surface such that $\Delta(x) > 0$ for all $x \in S$. Then*

$$C_1 = \frac{1}{2\pi} \int_S F_{ab} dx^a \wedge dx^b \in \mathbb{Z}. \quad (8)$$

Proof. By Theorem 1, the ground state $|\psi(x)\rangle$ is C^∞ on S . Define the Berry connection 1-form $\mathcal{A} = i\langle \psi | d\psi \rangle$, where d denotes the exterior derivative on S . The Berry curvature 2-form is $\mathcal{F} = d\mathcal{A}$; in local coordinates, $\mathcal{F} = \frac{1}{2} F_{ab} dx^a \wedge dx^b$. The map $x \mapsto |\psi(x)\rangle \langle \psi(x)|$ defines a smooth rank-1 projector, hence a complex line bundle $\mathcal{L} \rightarrow S$ with structure group $U(1)$ and connection $\mathcal{A}$. By the Chern–Weil theorem [12, 18], the first Chern number $C_1 = \frac{1}{2\pi} \int_S \mathcal{F}$ is an integer, since it equals the degree of the classifying map $S \rightarrow \mathbb{CP}^{n-1}$. $\square$

Caveat. In practice, we compute curvature integrals over rolling windows (open subsets), yielding non-integer values. Topological protection does not strictly apply.

Theorem 4 (Curvature–gap bound). *For all $x \in U$ and indices a, b ,*

$$|F_{ab}(x)| \leq \frac{2 \|\partial_a H(x)\|_{\text{op}} \|\partial_b H(x)\|_{\text{op}}}{\Delta(x)^2}, \quad (9)$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm. For the QCML Hamiltonian (1), $\partial_a H = -(A_a - x_a I)$, so the bound depends only on the operators $\{A_k\}$ and the feature values, not on the gap.

Proof. First-order perturbation theory gives the exact representation

$$F_{ab} = -2 \text{Im} \sum_{n \geq 1} \frac{\langle \psi_0 | \partial_a H | \psi_n \rangle \langle \psi_n | \partial_b H | \psi_0 \rangle}{(E_n - E_0)^2}, \quad (10)$$

where $\{|\psi_n\rangle\}_{n \geq 0}$ is a complete eigenbasis of $H(x)$. Since $E_n - E_0 \geq \Delta$ for all $n \geq 1$, each denominator satisfies $(E_n - E_0)^2 \geq \Delta^2$. Hence

$$|F_{ab}| \leq \frac{2}{\Delta^2} \sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle|.$$

By the Cauchy–Schwarz inequality on the sum,

$$\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle| \leq \left(\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \right)^{1/2} \left(\sum_{n \geq 1} |\langle \psi_n | \partial_b H | \psi_0 \rangle|^2 \right)^{1/2}.$$

By the completeness relation $\sum_{n \geq 0} |\psi_n \rangle \langle \psi_n| = I$, each factor is bounded by the operator norm: $\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \leq \|\partial_a H | \psi_0 \rangle\|^2 \leq \|\partial_a H\|_{\text{op}}^2$. Combining yields (9). $\square$

Proposition 5 (QFI volume). *Let g be a $p \times p$ positive semi-definite matrix of rank $r \leq p$, with positive eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > 0$. Define the pseudo-determinant*

$$\det^+(g) = \prod_{i=1}^r \lambda_i. \quad (11)$$

Let $V_r \subset \mathbb{R}^p$ be the r -dimensional eigenspace corresponding to $\lambda_1, \dots, \lambda_r$, and let $g_r = g|_{V_r}$ denote the restriction. Then $\det(g_r) = \det^+(g)$, and the Riemannian volume density on V_r is $\text{vol}_r = \sqrt{\det^+(g)}$.

Proof. By the spectral theorem, g admits an orthonormal eigenbasis $\{e_1, \dots, e_p\}$ with $g e_i = \lambda_i e_i$, where $\lambda_i > 0$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$. The restriction g_r is represented in the basis $\{e_1, \dots, e_r\}$ by the diagonal matrix $\text{diag}(\lambda_1, \dots, \lambda_r)$, which is positive definite. Therefore $\det(g_r) = \prod_{i=1}^r \lambda_i = \det^+(g)$. The Riemannian volume density on the r -dimensional submanifold equipped with metric g_r is $\text{vol}_r = \sqrt{\det(g_r)} = \sqrt{\det^+(g)}$, the standard formula for the volume element of a Riemannian metric [15]. $\square$

B Granger Causality

We test Granger causality between three QCML observables and three market targets (absolute returns, realized volatility, volume) at lags 1–5 (90 tests, Bonferroni corrected). QFI Determinant Granger-causes absolute returns at lag 1 ($F = 9.5$, $p = 0.0004$, Bonferroni-significant). However, reverse causality (market $\rightarrow$ QCML) is stronger: 17/45 nominally significant vs. 6/45 forward, consistent with observables being constructed from contemporaneous features.

C Default Hyperparameters

Table 12: Default hyperparameters.

Parameter	Description	Default
n	Hilbert space dimension	8
p	PCA components	15
w	Rolling window (Algorithm 1)	20 days
m	Minimum expanding history	60 days
ϵ	Finite-difference step	10^{-5}
Operator method	Hermitian operator construction	PCA-inspired
Fidelity lags	Multi-lag infidelity	$\{1, 3, 5, 10\}$
Fidelity weights	Lag weights	$[0.4, 0.3, 0.2, 0.1]$

D Numerical Stability Ablation

We sweep finite-difference $\epsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ and PCA dimension $p \in \{5, 10, 15, 30\}$ on four representative crises (GFC, Flash Crash, COVID, SVB) with fixed hyperparameters (no per-crisis tuning). Results appear in Table 13. Berry curvature is extremely stable across ϵ (all median $d \approx 0.15$), while QFI determinant shows moderate sensitivity ($d = 0.52$ at $\epsilon = 10^{-3}$ vs. $d = 0.36$ at 10^{-5}), saturating below 10^{-5} . Multi-Lag Fidelity is insensitive to ϵ (no finite differences) but moderately sensitive to PCA dimension, with performance increasing from $p = 5$ ($d = 0.29$) to $p = 10$ ($d = 0.44$) and stabilizing thereafter.

Table 13: Numerical stability: median Cohen’s d across 4 crises. Multi-Lag Fidelity does not use ϵ (no finite differences).

Config	Berry	QFI Det.	Multi-Lag
$\epsilon = 10^{-3}$	0.15	0.52	—
$\epsilon = 10^{-4}$	0.15	0.60	—
$\epsilon = 10^{-5}$ (default)	0.15	0.36	—
$\epsilon = 10^{-6}$	0.15	0.36	—
$p = 5$	0.09	0.34	0.29
$p = 10$	0.20	0.41	0.44
$p = 15$ (default)	0.15	0.36	0.45
$p = 30$	0.22	0.54	0.45

E Window-Size Sensitivity

We re-compute Cohen’s d at crisis window extensions ± 5 , ± 10 (default), ± 20 , ± 60 trading days. Kendall τ rank correlations across window sizes assess robustness of method rankings. Results appear in Table 14. Rankings are stable for nearby windows ($\tau = 0.87$ for ± 5 vs. ± 10 and ± 5 vs. ± 20) but degrade at ± 60 days ($\tau = 0.07$ – 0.33), where diluted crisis signal reduces separation. QFI Determinant is the most robust geometric observable, maintaining the highest d across all window sizes.

Table 14: Median Cohen’s d by crisis window size (12 crises).

Method	$\pm 5d$	$\pm 10d$	$\pm 20d$	$\pm 60d$
Berry Curv. Incr.	0.25	0.29	0.27	0.24
QFI Determinant	0.63	0.36	0.64	0.43
Multi-Lag Fidelity	0.40	0.27	0.38	0.26
Random Forest	0.21	0.21	0.28	0.34

F Hyperparameter Sensitivity: Fixed vs. Tuned

Table 15 compares three hyperparameter regimes: (a) fixed defaults with all `pca_inspired` operators, (b) fixed defaults with per-method operator ablation (Berry uses `random`; QFI and MLF use `pca_inspired`—these are the main results in Table 4), and (c) Optuna HPO (100 trials, pre-2020 training crises, post-2020 validation). The operator ablation provides a modest improvement for

Berry (+4%) with larger changes for QFI and MLF that reflect pipeline-level differences requiring further investigation. Naïve HPO showed severe overfitting (train d exceeding validation by 0.35–0.75); regularized HPO with a consistency penalty reduces this gap to ≤ 0.20 , with QFI achieving val $d = 0.60 > \text{train } d = 0.50$ (column c*).

Table 15: Hyperparameter sensitivity: median Cohen’s d across 12 crises (columns a–b) and train/val split (column c).

Method	(a) All PCA-Insp.	(b) Per-Method Op.	(c) HPO Val.
Berry Curvature Increment	0.28	0.29	0.49
QFI Determinant	0.68	0.36	0.60
Multi-Lag Fidelity	0.36	0.27	0.33

(a,b) Fixed $n = 8$, $p = 15$, $w = 20$. (c*) Regularized Optuna TPE, 100 trials; consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$; tighter search space; operator method fixed by ablation; validation on 4 post-2020 crises. Train d : Berry 0.68, QFI 0.50, MLF 0.53. Train–val gap ≤ 0.20 (vs. 0.35–0.75 without regularization).